

Polarization of Light

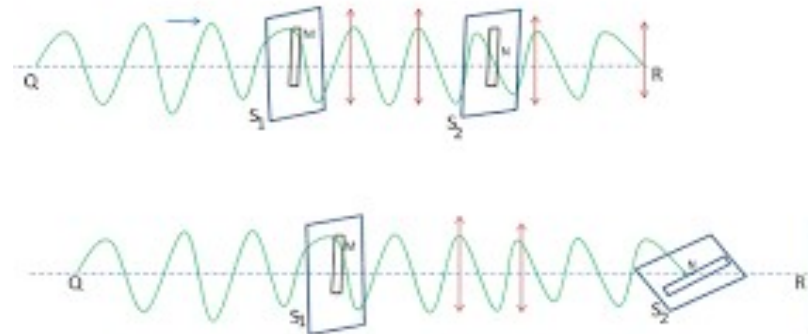
Watch this introductory video

<https://www.youtube.com/watch?v=DjnDX28l4xA>

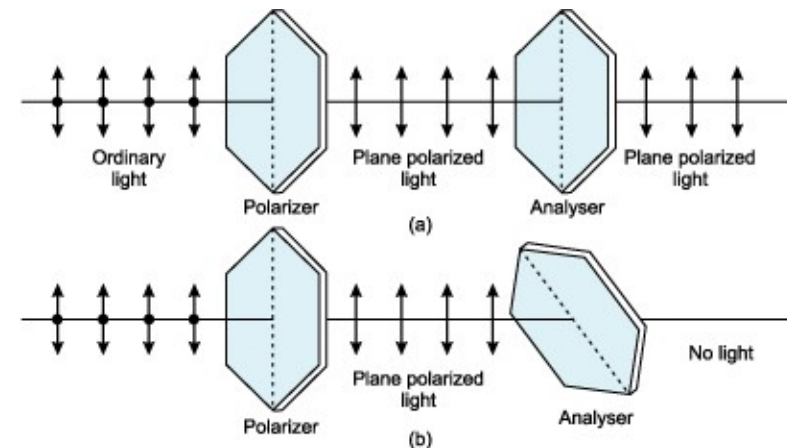
Polarization of Light

- Polarization exhibits transverse nature of light or electro-magnetic wave.
- The light having vibrations along all possible directions is known as unpolarized light.
- The light vibrations only along a single straight line perpendicular to the direction of light is said to be polarized.
- A light wave is said to be plane or linearly polarized if the electric field vector or magnetic field vector at any point goes on vibrating along the same line perpendicular to the direction of propagation of light wave.

Polarization in string-slit experiment (Transverse wave)



Polarization of light through optical crystals

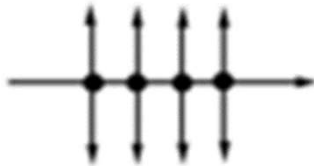


a) Unpolarised Light:



b) Unpolarised Light:

In terms of resolved components

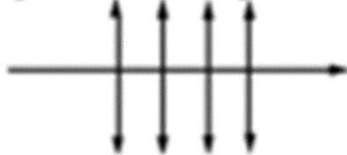


c) Partially Polarised Light



d) Plane Polarisised Light,

plane polarised in the plane of incidence



e) Plane Polarisised Light

plane polarised in the plane perpendicular to the plane of incidence

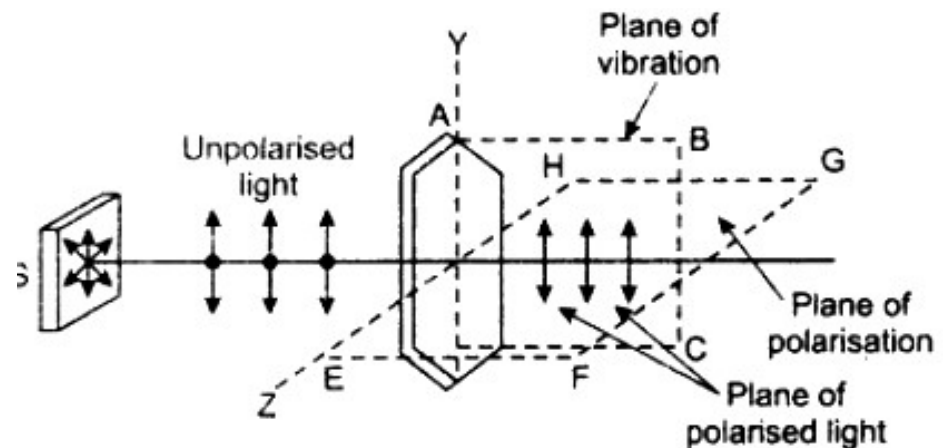
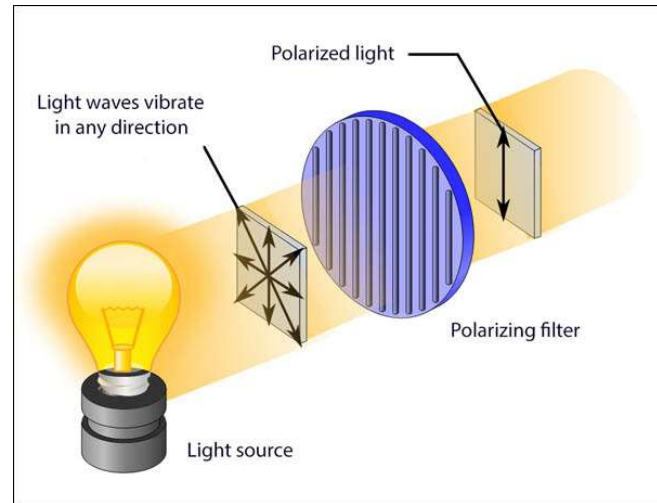


Fig : Plane of vibration and plane of polarisation

Methods to get plane polarized light

- **Polarization by reflection**
- Polarization by refraction
- **Polarization by double refraction**
- Polarization by scattering
- Polarization by selective absorption

Polarization by reflection: Brewster's Law

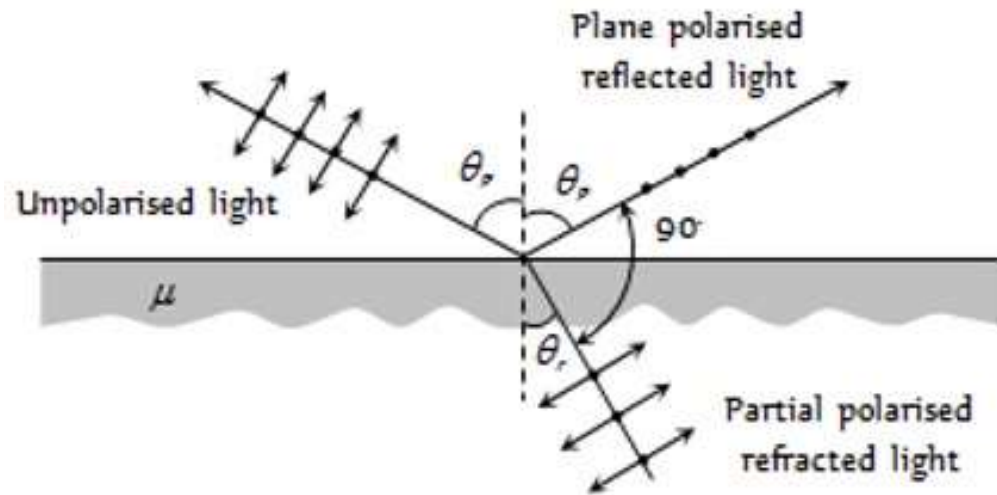
In 1881, Sir David Brewster performed experiments to observe polarization by reflection. He observed that,

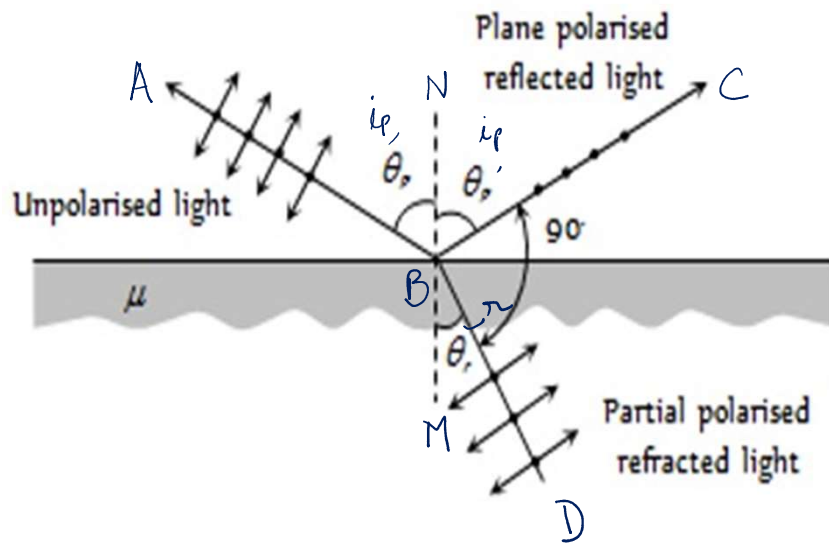
When a beam of unpolarized light is incident at a surface of a transparent medium, the reflected ray will be completely plane polarized for a specific angle of incidence known as polarizing angle or Brewster angle. The refracted ray will be partially polarised.

Brewster discovered the relation,

$$\mu = \tan \theta_p$$

The tangent of angle of polarization (θ_p) is equal to the refractive index (μ) of the transparent medium. This is called Brewster's law.





At polarising angle, the reflected ray and refracted ray are perpendicular to each other

From, Brewster's law, $\mu = \tan i_p = \frac{\sin i_p}{\cos i_p}$ — (1)

From Snell's law, $\mu = \frac{\sin i_p}{\sin r}$ — (2)

Equating R.H.S. of (1) & (2),

$$\frac{\sin i_p}{\cos i_p} = \frac{\sin i_p}{\sin r}$$

$$\text{or, } \cos i_p = \sin r = \cos\left(\frac{\pi}{2} - r\right)$$

$$\angle i_p = \frac{\pi}{2} - \angle r$$

$$\angle i_p + \angle r = \pi/2 \quad \text{--- (3)}$$

$$\text{In } \angle NBM, \angle i_p + \angle CBD + \angle r = \pi$$

$$\angle CBD = \pi - \pi/2 = \pi/2$$

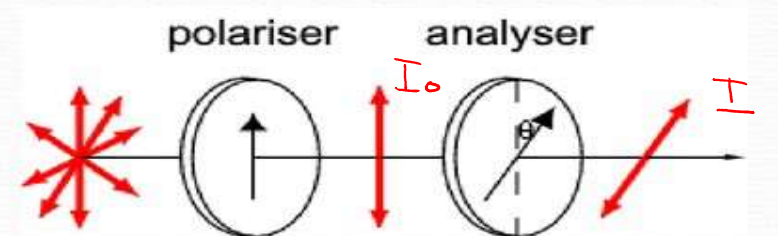
$$\boxed{\angle CBD = \pi/2}$$

Malus' Law

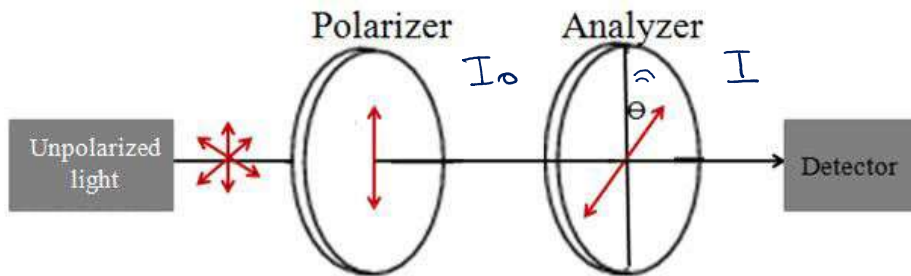
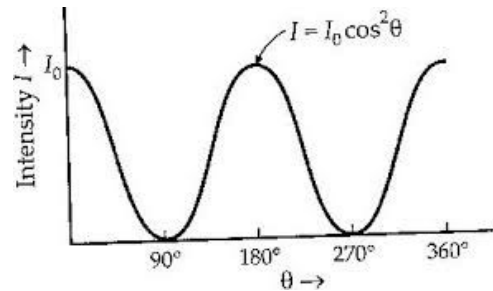
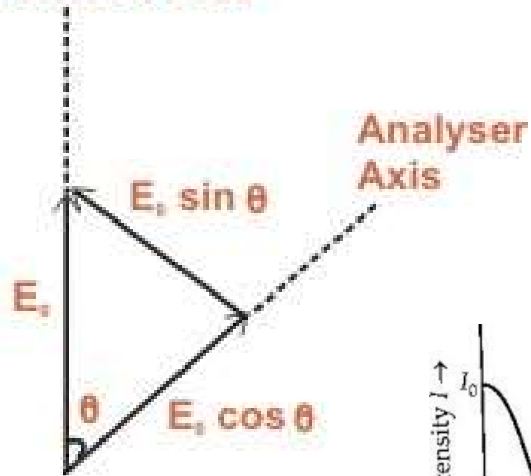
- This is the *law of Malus* named after Etienne Malus who published this relationship in 1809.
- According to Malus, - *when completely plane polarized light is incident on the analyzer, the intensity (I) of the light transmitted by the analyzer is directly proportional to the square of the cosine of angle between the transmission axes of the analyzer and the polarizer.*

- i.e $I \propto \cos^2 \theta$

$$I = I_0 \cos^2 \theta$$



Polariser Axis



If E_0 is the amplitude of plane polarised light.

$\Rightarrow$ At analyser, E_0 is divided into two components. $E_0 \sin \theta$ & $E_0 \cos \theta$

$\Rightarrow E_0 \sin \theta$ is absorbed at analyser, however $E_0 \cos \theta$ is passed through the analyser which is along the transmission axis of analyser.

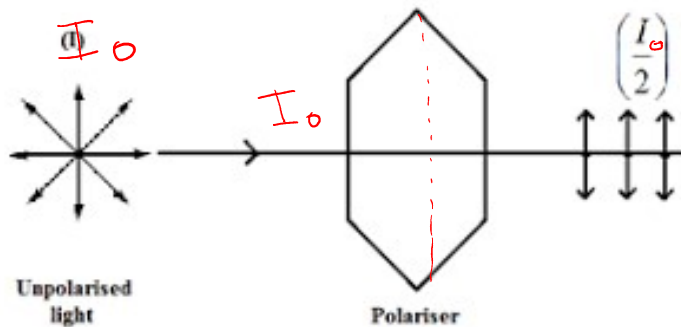
Amplitude of PPL after analyser

$$E' = E_0 \cos \theta$$

Intensity, $I = E'^2 = E_0^2 \cos^2 \theta$

$$\boxed{I = I_0 \cos^2 \theta}$$

- Malus law does not hold when the incident light is unpolarized.
- For unpolarized light, the angle ' θ ' is not constant and can have random values.
- Hence, for such case, In Malus law, we have to consider average value of $\cos^2(\theta)$ over all possible values of θ , which is $\frac{1}{2}$. $[\overline{\cos^2 \theta} = \frac{1}{2}]$
- Hence an ideal polariser is one that transmits 50% of the incident unpolarized light as plane polarized light.



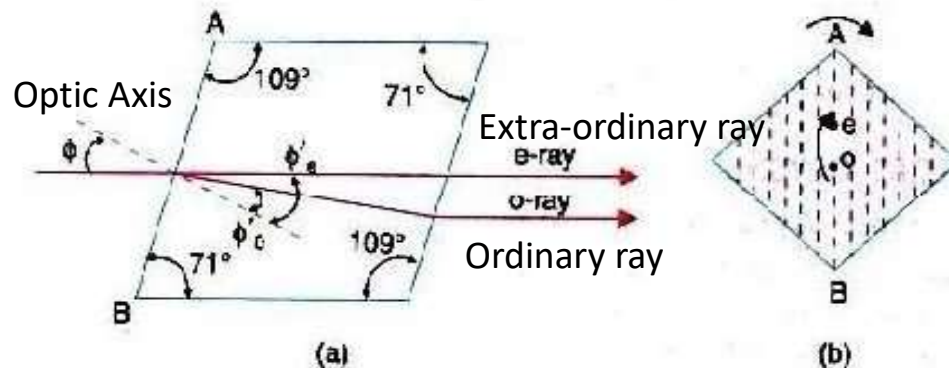
$$I = I_0 \cdot \cos^2 \theta = I_0/2$$

$$I = I_0/2$$

Here, $I_0 \rightarrow$ Intensity of unpolarized light
 $I \rightarrow$ " " " polarised light

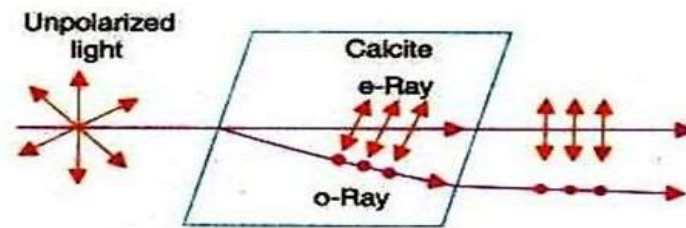
Double refraction

□ When a ray of light is refracted by a crystal of calcite it gives two refracted rays. This phenomenon is called "**DOUBLE REFRACTION**".



If we put calcite crystal over a dot mark on the paper, two images of dot are observed from top of crystal. Now, on rotating the crystal, one image remains fixed (corresponding to O-Ray) however one rotating with crystal (corresponding to E-Ray). Crystal is isotropic for O-Ray and anisotropic for E-Ray.

Double refraction



Ordinary ray travels with same velocity in all the direction inside the crystal.

Extra-ordinary ray travel with different velocities in different directions inside crystal.

- The two rays produced in double refraction are linearly polarized in mutually perpendicular directions.
- The ray which obeys Snell's law of refraction is known as **ordinary ray** or **o-ray**.
- The other ray does not obey Snell's law is called **extraordinary ray** or **e-ray**.

Optic Axis: An optic axis of a crystal is a direction in which a ray of transmitted light suffers no double refraction. An optical axis is a direction rather than a single line: all rays that are parallel to that direction exhibit the same lack of birefringence. Along optic axis, both the ordinary and extraordinary ray travel with same velocity inside the crystal.

As per Huygens's theory, When a ray of unpolarized light is incident on doubly refracting crystal, it splits into two different wavefronts that travel inside the crystal.

One wavefront is spherical that forms the Ordinary ray that follows law of refraction. Another is Ellipsoid wavefront that forms the Extra-ordinary ray that does not follow law of refraction.

The ellipsoid of revolution is obtained by rotating an ellipse about either major or minor axis of an ellipse. This axis of rotation is along the optic axis.

If ellipsoid lies completely outside the spherical wavefront, then the crystal is known as **negative crystal**. In negative crystal velocity of Extra-ordinary ray remains more than ordinary ray inside the crystal. Ex. Calcite.

If ellipsoid lies completely inside the spherical wavefront, then the crystal is known as **positive crystal**. In positive crystal velocity of Ordinary ray remains more than Extra-ordinary ray inside the crystal. Ex. Quartz.

